

Supplementary Information

Reproducible Radial–Angular Geometry of Visual Category Representations in Human MEG

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1 Notation and pre-specified primary analysis

The main manuscript defines the full analysis notation. Briefly, D denotes a category-to-category distance and R denotes cross-run reliability of the resulting six-distance representational dissimilarity matrix (RDM). FULL is the full complex Euclidean distance, AMP is its amplitude-difference component, and AP is its amplitude-weighted angular component. Reliability R ranges from -1 to 1 because the primary association measure is Spearman rank correlation.

The prospective confirmation analysis used 102 magnetometers, $K = 16$ graph Fourier transform modes, theta 4–8 Hz and alpha 8–13 Hz, the 50–250 ms poststimulus window, rank-2 category-independent common-subspace removal, alpha modal-power normalization, all 15 disjoint 2-vs-2 partitions of five runs, and the exact FULL/AP/AMP distance definitions. These choices were fixed before inspection of confirmation-cohort neural outcomes. The same representation was then applied unchanged to a second non-overlapping cohort from the same COGITATE resource. Later component and phase-null analyses reused these participants as robustness and mechanism-oriented analyses; they are not additional independent participant replications.

2 Exact component identities

For category responses $z_i = A_i e^{i\phi_i}$ and $z_j = A_j e^{i\phi_j}$, with $\Delta\phi_m = \phi_{i,m} - \phi_{j,m}$,

$$D_{\text{FULL}} = \sum_m |z_{i,m} - z_{j,m}|^2, \quad (1)$$

$$D_{\text{AMP}} = \sum_m (A_{i,m} - A_{j,m})^2, \quad (2)$$

$$D_{\text{AP}} = \sum_m 2A_{i,m}A_{j,m}[1 - \cos(\Delta\phi_m)], \quad (3)$$

$$D_{\text{FULL}} = D_{\text{AMP}} + D_{\text{AP}}. \quad (4)$$

The same full distance can be written as

$$D_{\text{ENERGY}} = \sum_m (A_{i,m}^2 + A_{j,m}^2), \quad (5)$$

$$D_{\text{PRODUCT}} = \sum_m 2A_{i,m}A_{j,m}, \quad (6)$$

$$D_{\text{INTERFERENCE}} = - \sum_m 2A_{i,m}A_{j,m} \cos(\Delta\phi_m), \quad (7)$$

$$D_{\text{ANGLE}} = \sum_m 2[1 - \cos(\Delta\phi_m)], \quad (8)$$

18 with

$$D_{\text{AMP}} = D_{\text{ENERGY}} - D_{\text{PRODUCT}}, \quad (9)$$

$$D_{\text{AP}} = D_{\text{PRODUCT}} + D_{\text{INTERFERENCE}}, \quad (10)$$

$$D_{\text{FULL}} = D_{\text{ENERGY}} + D_{\text{INTERFERENCE}}. \quad (11)$$

19 ANGLE is an amplitude-free angular diagnostic; INTERFERENCE is a signed algebraic component
20 and is not treated as a distance metric by itself.

21 3 Core cohort results

Table 1: Cross-run RDM reliabilities. Confidence intervals are descriptive participant-bootstrap 95% intervals based on 100,000 resamples.

Endpoint	Confirmation cohort	Generalization cohort
FULL	0.785 [0.716,0.839]	0.680 [0.603,0.749]
AP	0.748 [0.680,0.804]	0.654 [0.575,0.725]
AMP	0.606 [0.500,0.701]	0.485 [0.375,0.589]
AP-AMP	0.142 [0.081,0.210]	0.169 [0.116,0.225]

22 The prospective AP-AMP test yielded one-sided participant sign-flip $p = 7 \times 10^{-5}$. The confirmation
23 FULL endpoint had 30/30 participants positive and no exceedance among 100,000 sign flips ($p < 10^{-5}$).

24 4 Coordinate, temporal, and association-metric robustness

Table 2: Selected robustness results. OOB-PCA means out-of-bag principal component analysis, in which the unused fifth run fits the PCA basis for each 2-vs-2 partition. Kendall τ_b is a rank-association coefficient that corrects for ties.

Control	Confirmation cohort	Generalization cohort
OOB-PCA AP-AMP	0.102 ($p = 2.1 \times 10^{-4}$)	0.166 ($p < 10^{-5}$)
Rank-matched sensor AP-AMP	0.143 ($p = 1.1 \times 10^{-4}$)	0.116 ($p < 10^{-5}$)
Random rotations: median AP-AMP	0.149	0.166
Random rotations with positive contrast	49/49	49/49
Far-prestimulus FULL	-0.043	0.002
Poststimulus FULL	0.785	0.680
Pearson AP-AMP	0.132	0.163
Kendall τ_b AP-AMP	0.130	0.133

25 These controls establish robustness within the tested low-dimensional sensor-derived subspace. They
26 do not establish anatomical source-space generality.

27 5 Radial and angular component reliability

28 PRODUCT was more reliable than AP by 0.0563 in the generalization cohort ($p = 0.0077$), supporting
29 a mixed radial-angular interpretation rather than phase dominance.

Table 3: Exact component reliability, mean [95% confidence interval].

Component	Confirmation cohort	Generalization cohort
ENERGY	0.790 [0.720,0.844]	0.719 [0.656,0.774]
PRODUCT	0.770 [0.697,0.829]	0.710 [0.646,0.767]
INTERFERENCE	0.743 [0.663,0.813]	0.588 [0.488,0.682]
ANGLE	0.684 [0.593,0.762]	0.495 [0.397,0.588]
INTERFERENCE→FULL	0.746 [0.677,0.806]	0.618 [0.533,0.696]

6 Amplitude-preserving category-phase null

The phase-specific null leaves amplitudes assigned to their original categories and randomly permutes only the four category phase labels at each graph-mode/time coordinate, independently between compared run halves. It therefore preserves the amplitude field and the coordinate-wise set of phase values while destroying category-specific phase assignment.

Table 4: Observed component reliability versus amplitude-preserving phase-label null.

Cohort	Component	Observed	Null	Difference	95% CI	p
Confirmation	INTERFERENCE	0.743	0.531	0.212	[0.162,0.262]	$< 10^{-5}$
Confirmation	ANGLE	0.684	-0.001	0.685	[0.597,0.762]	$< 10^{-5}$
Generalization	INTERFERENCE	0.588	0.483	0.105	[0.024,0.188]	0.0111
Generalization	ANGLE	0.495	-0.001	0.497	[0.398,0.590]	$< 10^{-5}$

ANGLE observed-minus-null effects were positive in 29/30 participants in each cohort. Implementation audits reconstructed the pre-null component scores to maximum absolute difference 1.11×10^{-16} ; amplitude preservation error was at most 2.22×10^{-16} ; phase-multiset audit error was zero.

7 The six-distance RDM and its limitation

With categories face (F), object (O), letter (L), and false font (X), each RDM contains exactly six unique distances: $F-O$, $F-L$, $F-X$, $O-L$, $O-X$, and $L-X$. For each participant, the 15 overlapping 2-vs-2 run partitions are averaged into one reliability score before population inference, avoiding partition-level pseudoreplication. Nevertheless, six distances are intrinsically low-dimensional. The archived confirmatory exports retain participant-level aggregate reliabilities rather than the complete pair-specific splitwise vectors needed for a valid post hoc item-level stability analysis. No claim is therefore made that all six category pairs contribute equally. Future reruns should retain pair-level splitwise values, report leave-one-pair-out sensitivity, and use richer stimulus sets to produce denser RDMs.

8 Ensemble-size dependence

Mean FULL reliability was approximately 0.008, 0.021, 0.075, 0.207, 0.404, 0.624, and 0.757 when category fields were estimated from 1, 2, 4, 8, 16, 32, and 64 repeated trials, respectively. A descriptive saturation fit gave $R_\infty = 0.847$ and a trial-count scale $\tau_n = 26.36$. These values describe the observed convergence curve; τ_n is neither a neuronal time constant nor a prescriptive trial requirement.

9 Standardized participant-level effect sizes

As an additional descriptive scale, paired standardized mean differences (Cohen’s d_z , mean paired effect divided by the standard deviation of paired effects) were computed from archived participant-level values. For AP-AMP, $d_z = 0.77$ in the confirmation cohort and $d_z = 1.08$ in the within-resource generalization cohort. For the phase-null comparison, ANGLE observed-minus-null yielded $d_z = 2.89$ and 1.82 , whereas INTERFERENCE observed-minus-null yielded $d_z = 1.50$ and 0.44 in the two cohorts, respectively. These standardized values are descriptive additions and do not replace the pre-specified sign-flip inference.

10 Interpretation boundary

The evidence supports reproducible mixed radial-angular visual-category geometry in a defined early-alpha MEG representation. The effect is demonstrated in a sensor-derived 16-dimensional subspace; anatomical source-space generality, independent-dataset replication, and causal perturbation remain future tests.